

for lacking tokens bearing the Res feature (they are evenly split between experientials and universals), a feature which is born by 83 of 115 (72.2%) tokens in the data set. This may suggest that the Chinese TAM system makes use of a +Res/-Res contrast in which *you* is never associated with +Res, though more data and testing are required to reinforce this hypothesis.

The *suo...de* → [varies] error is likewise noteworthy for appearing only in the MLP experiment data, and also for the fact that every token in the set is an experiential such that it represents 55.9% of all Exp prediction errors (i.e. errors that bear the Exp feature). Taken in tandem with a quite low *p*-value for the Exp feature in the Chinese MLP

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data set, this all suggests that experientiality is highly salient to this model, though any specific claims must remain judicious in light of this feature's relatively low ϕ -value.

Similarly, while the *le* → *unmarked* and *le* → *suo...de* errors suggest a correlation with Aktionsart (accomplishment and achievement specifically), the Aktionsart *p*-values for both models did not indicate feature significance. It is possible that further testing involving a larger data set and a refined feature schema may illuminate the role of Aktionsart in the Chinese TAM system.

While not presented in depth here, the adjoined data tables show that the *unmarked* → *le* (RFC) and *unmarked* → (varies) (MLP) all exhibit relatively random feature distribution. It is potentially interesting in and of itself that the two models encountered different error patterns here; this observation can inform how the feature set is developed, though exactly *how* is still an open question⁹.

Speaking generally to both models' performance on resultatives (better than baseline) and experientials (worse than baseline), it bears noting that these features are in nearly complementary distribution in the data set, the exceptions being the 15/300 tokens that are ambiguous Exp/Res or 3/115 in the Chinese experiment specifically. This may indicate that rather than distinguishing between resultatives and experiences, the models are distinguishing between resultatives, which they handle well, and everything else, a group within which they are unable to derive nuanced distinctions based on the current feature set.

From a purely linguistic perspective, the results of both experiments suggest that the current feature schema is inadequate to the task of mapping the Chinese PLC landscape. One likely reason for this is that, as noted above, the majority of the research which has been conducted on PLCs has focused on Indo-European, and especially



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This should be easy to tabulate and present here?



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What's the baseline?



Cross-Linguistic Patterns and Insights

- 300 tokens in 3 languages is a great number for a pilot experiment, but much more data will be required for truly compelling insights; but that's okay — that's what the “real” (non-pilot) experiments are there for
- Error patterns between English and German were fairly similar, while the error patterns with Chinese were wholly distinct
 - This in some ways poses problems for the methodology — can **perfects/PLCs even be considered a coherent class?** — on the other hand it's extremely useful in that more (and better annotated by actual L1 speakers) data may reinforce some of the patterns that here are suggestive but not as yet truly compelling
 - This is totally expected: perfects/PLCs are notoriously “slippery” in their semantics, not only cross-linguistically but even intra-linguistically
- Unfortunately, there aren't *that* many real standouts. That said, this is a pilot experiment with a small sample size (300 tokens in three languages with a train/test split of 50/50), the features are preliminary and as such somewhat naive (they're pulled from the literature and identified by rough frequency of mention — i.e. these are some of the most commonly discussed/proposed semantic features of cross-linguistic PLCs), and were always going to need significant refinement. I just wish the signals as to *how* to refine them **were a weeeeeee bit clearer...**
- With respect to Chinese, both models showed a pattern of performing particularly well on resultatives and particularly poorly on experientials, even though the strength of association for both features and both models is relatively weak
 - The fact the same pattern appeared in both, however strong the association, suggests that these two features are significant for how the models interpret the data
 - From a linguistic perspective, this may mean, for example, that one particular construction is being associated (correctly in the case of resultatives, incorrectly in the case of experientials) with each of these features
 - Based on the error patterns seen in the previous section, it appears that the model defaults to *le* for resultatives, even though *suo_de* and *unmarked* also occur frequently with resultatives
 - *For analysis in the thesis: what about the Chinese TAM landscape affects this pattern?*
 - There is nothing in the error patterns that points specifically to the Exp feature; it's more or less evenly spread across error types. My read is that the Chinese TAM system is moderated by (is this the right wording? “operates on”? “is conditioned by”? “hinges on”?) a different featural “constellation” than English and German, which despite their grammatical differences, are in fact still closely related languages that have long existed in near contact with each other
 - While not picked up as significant in the X-square analysis, it is noteworthy that for both models, the *le* → *suo_de* error correlates strongly with accomplishments, and the *le* → *unmarked* error correlates just as strongly with achievements
- With respect to German, there are a few significant observations that can be made: both models show significant (by X-square analysis) features, though most of them have relatively weak ϕ -values; both models



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Dahl and Wälchli



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Well, start by splitting the predicted categories — that is my main hang up in general. If I'm being harsh, I would say the results on these very fine-grained categories just don't say much.

